Numerical Analysis

Lab 8

Giuseppe Caliò

A.Y. 2024-2025

November 18,2024

Polynomial interpolation

This lab deals with polynomial and piecewise polynomial interpolation.

Table of contents

Polynomial interpolation I

Given a function $f(x)$ the **interpolating function** in a finite dimensional space $V_k \subseteq V$ is

$$g(x) = \sum_{i=0}^{n} \alpha_i \phi_i(x) \qquad \text{s.t.} \qquad g(x_i) = f(x_i) \quad i = 0, 1, \dots, n$$

where $\{x_i\}_{i=0}^{n}$ and $\{\phi_i(x)\}_{i=0}^{n}$ are respectively sets of $k+1$ **nodes** and **basis functions**; the coefficients α_i are called **weights**.

Definition 8.1 (Lagrange interpolating polynomial)

Lagrange polynomials are defined taking as set of basis functions

$$\varphi_j(x) = \prod_{j=0, i \neq j} \frac{(x - x_i)}{(x_j - x_i)}$$

and they are s.t. $\varphi_j(x_i) = \delta_{ij}$. The weights are $\alpha_i = f(x_i)$ and the Lagrange interpolating polynomial is

$$\Pi_n f(x) = \sum_{i=0}^{k} f(x_i) \varphi_i(x).$$

Polynomial interpolation II

Theorem 8.1 (Interpolation error)

Let $\Pi_n f(x)$ be the interpolating polynomial of order n at the nodes $x_i \in [a, b]$. If $f(x) \in \mathcal{C}^{n+1}([a, b])$ then

$$E_n f(x) = f(x) - \Pi_n f(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i) \qquad \xi \in [a, b],\ \forall x \in [a, b].$$

If the nodes are equally spaced with step h, it holds

$$\| E_n f(x) \|_{L^\infty([a,b])} \leq \frac{\| f^{(n+1)}(x) \|_{L^\infty([a,b])}}{4(n+1)} h^{n+1}$$

This does not ensure a priori the uniform convergence of $\Pi_n f(x)$ to $f(x)$ for $n \to \infty$.

Polynomial interpolation III

Exercise 8.1

Consider the following points

$$P_1\,(0,\,5),\;\; P_2\,(1,\,3),\;\; P_3\,(3,\,5) \quad \text{and} \quad P_4\,(4,\,12).$$

a. Find the interpolation polynomial of minimum order that interpolates such points using Lagrange polynomials as basis functions.
b. Plot the values attained by the obtained polynomial on a finer grid of nodes and verify that the the polynomial actually interpolates the points.
c. Repeat the operation using MATLAB command `polyfit`.

Exercise 8.2

Consider the problem of approximating the function $f(x) = \sin(x)$ in the interval $[0, \pi]$ with a polynomial of degree 9.

a. Consider a set of equally spaced nodes x_i and find the interpolating polynomial using Lagrange basis polynomials.
b. Find again the polynomial using `polyfit` command.
c. Estimate the interpolation error using the related theorem. Then evaluate the interpolation polynomial in 1000 equally spaced points in the interval and plot the interpolation error. Motivate the result.
d. Repeat the operations with polynomials of order $k = 1, \ldots, 30$. Plot the behaviour of $\max_x |e_k(x)|$ as a function of k and comment the results.

Polynomial interpolation V

Exercise 8.3 (Runge's counterexample)

Consider the function

$$f(x) = \frac{1}{1+(5x)^2} \qquad -1 \leq x \leq 1.$$

- (a) Verify that the interpolation polynomials $p_k(x)$ using equally spaced nodes are such that $$\lim_{k\to\infty} |f(x) - p_k(x)| \neq 0.$$
- (b) Overcome this problem using non uniformly distributed nodes, e.g. **Chebyshev nodes** which in $[-1, 1]$ are

$$x_i = \cos\left(\frac{1+2i}{2(k+1)}\pi\right) \qquad i = 0, 1, \ldots, k.$$

and can be mapped to a generic interval $[a, b]$ with the transformation

$$t_i = \frac{b-a}{2}x_i + \frac{a+b}{2}.$$

Polynomial interpolation VI

Notice that the commands polyfit and polyval can be used to compute both the interpolation and a least-square approximation

Note

polyfit(x_{nodes}, y_{nodes}, N)

$$
\begin{aligned}
N+1 &= \#x\_nodes \implies \text{Lagrange interpolation;} \\
N+1 &!= \#x\_nodes \implies \text{least square interpolation.}
\end{aligned}
$$

Table of contents

Piecewise polynomial interpolation I

When using n equally spaced nodes, uniform convergence of the interpolating polynomial is not guaranteed for $n \to \infty$.

Definition 8.2 (Piecewise linear interpolation)

Given of nodes: $x_0 < x_1 < \cdots < x_n$, we denote by I_i the interval $[x_i, x_{i+1}]$ and by H the maximum length of these intervals.
We denote by $\Pi_1^H f$ the piecewise linear interpolating polynomial of f given by:

$$\Pi_1^H f(x) = f(x_i) + \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}(x - x_i) \quad \forall x \in I_i.$$

Piecewise polynomial interpolation II

Theorem 8.2 (Interpolation error)

If $f \in C^2(I)$, where $I = [x_0, x_n]$, then

$$\|f(x) - \Pi_1^H f(x)\|_{L^\infty(I)} \leqslant \frac{H^2}{8} \|f''(x)\|_{L^\infty(I)}$$

This bound allows us to ensure the uniform convergence of the interpolating polynomial to f.

Piecewise polynomial interpolation III

Exercise 8.4

a. Consider $f(x) = \sin(x)$ in the interval $[0, \pi]$: compute the interpolation with $n = 1, 2, 4, 8, 16, 32$ elements, find the error in infinity norm and verify it converges quadratically with the distance between two nodes h.
b. Consider now the Runge function

$$f(x) = \frac{1}{1 + (5x)^2} \qquad -1 \le x \le 1 :$$

compute the interpolation with $n = 1, 2, 4, 8, 16, 32$ elements, find the error in infinity norm and verify it converges quadratically with the distance between two nodes h.
c. Interpolate the Runge function using a piecewise cubic spline (`spline` command) with $n = 1, 2, 4, 8, 16, 32$ subintervals. Plot the result.

Piecewise polynomial interpolation IV

Exercise 8.5

Consider the function $f(x) = e^x$ in the interval $[-1, 1]$ and the problem of finding a piecewise linear interpolating polynomial on a set of nodes $\{x_i\}$.

a. Determine how many equally spaced nodes x_i are needed in order to get an interpolation error with infinity norm smaller than 10^{-3}.
b. Construct the interpolating polynomial using such number of nodes, compute the error numerically and check if the request is satisfied.

Exercise 8.6

Consider the function

$$f(x) = \left| x - \frac{\pi}{12} \right| \qquad -1 \leq x \leq 1.$$

a. Verify that the interpolating polynomials $\Pi_n f(x)$ based on equally spaced nodes exibit the Runge's phenomenon.
b. Compute the piecewise linear interpolation (`interp1` command) based on $n = 1, 2, 4, 8, 16, 32$ intervals and compute the associated error with the respect to infinity norm. Is the quadratic convergence ensured in such a case?
c. Interpolate the function using a piecewise cubic spline (`spline` command) with $n = 1, 2, 4, 8, 16, 32$ subintervals. Plot the result.

Table of contents

Homework 8.1

Consider the polynomial $p_4(x) = x^4$.

a. Approximate $p_4(x)$ with the cubic polynomial $p_3(x)$ interpolating $p_4(x)$ in $x = -1, 0, 1, 2$, using Lagrange polynomial as basis functions. Repeat the operation with the monomial basis.
b. Compute an estimate for $e(x) = \left|x^4 - p_3(x)\right|$.
c. Compute $e(x)$ and compare it with the estimate.

Homework II

Homework 8.2

Consider the function $f(x) = \log(2+x)$ in $[-1, 1]$ and its interpolating polynomial $p_n(x)$ in $n+1$ nodes.

a. Give an estimate of $\|f - p_n\|_\infty$ in case of both equally spaced and Chebyshev nodes.
b. Numerically verify the obtained results for $n = 2, 4, 8, 16$.

Homework 8.3

Consider the following function $f(x)$

$$f(x) = \frac{1}{(x-0.3)^2+0.01} + \frac{1}{x^2+0.04} - 6.$$

a. Sample $f(x)$ in 10 equally spaced nodes in the interval $[-1,3]$ and construct the minimum degree interpolating polynomial. Use Vandermonde matrix, Lagrange polynomials and `polyfit` commands.
b. Find the minimum degree polynomial interpolating $f(x)$ in the Chebyshev nodes in the same interval.
c. Numerically compute the error in both cases.

Homework IV

Homework 8.4

Let $f(x) = x^5 + 1$ and $x_i = 0.2 \cdot i$, with $i = 0, 1, 2, 3$.

a. Let $p(x)$ be the interpolation polynomial of $f(x)$ in the nodes x_i. Determine $p(x)$ and give an upper bound for $|e(x)| = |p(x) - f(x)|$.
b. Let $\overline{s}(x)$ be the linear spline interpolating $f(x)$ in the nodes x_i. Give an upper bound for $|\overline{e}(x)| = |\overline{s}(x) - f(x)|$.

Homework V

Homework 8.5

Consider the following function

$$f(x) = \begin{cases} 0 & x \in [0,1) \\ 1 & x \in [1,2]. \end{cases}$$

First interpolate the function using the Lagrange polynomials with an increasing number of nodes. Then repeat the procedure using linear splines. What do you observe? Justify the answer.

Homework 8.6

Approximate the function $f(x) = x\sin(x)$ in the interval $[-2, 6]$.

a. Consider a set of equally spaced nodes: find the interpolating polynomial using Lagrange basis polynomials (`polyfit`, `polyval` commands) of degree $4, 6, 8$. Plot the function $f(x)$ and the associate interpolating polynomials.
b. Estimate the interpolation error when resorting to a quadratic interpolation (n=2), using the theoretical results. Then evaluate the interpolating polynomial in 1000 equally spaced points and plot the interpolation error. Motivate the result.
c. Compute with Matlab the piecewise linear interpolant (`interp1` command) and the cubic spline (`spline` command) associated with $n + 1 = 6, 11, 21$ equally spaced nodes.

References I